

**NASDAQ INC** **NDQA** **S**October 02, 2018 - 12:46pm EST  
by

VIC\_Member2015

|                      |                    |          | 2018 | 2019 |
|----------------------|--------------------|----------|------|------|
| Price:               | 77.00              | EPS      | 3.5  | 3.9  |
| Shares Out. (in M):  | 312                | P/E      | 23   | 20   |
| Market Cap (in \$M): | 44M                | P/FCF    | 0    | 0    |
| Net Debt (in \$M):   | 0                  | EBIT     | 0    | 0    |
| TEV (in \$M):        | 44M                | TEV/EBIT | 0    | 0    |
| Borrow Cost:         | General Collateral |          |      |      |

## Description

This is an update to the recent write-up in light of new negative information that has come to light about the Exchanges.

We believe the US exchanges (i.e. CBOE, NDQ, ICE) are extraordinarily attractive short investments. Especially so in the near-term and when viewed from a risk/reward perspective.

### Summary

Our view is that the chances of regulatory reform of the U.S. Exchanges are quite high, a fact the market has failed to appreciate. As we describe below, (1) the Exchanges have begun raising prices on their clients for the past decade; a fact which on its face seems odd because their clients are some of the biggest companies in the world with significant market power – clients like Goldman Sachs, JPMorgan and Fidelity; (2) this has occurred due to a unique combination of antiquated exchange rules; significant consolidation among the exchanges (12 of the 13 exchanges in the US are now owned by just three companies) and regulations that have yet to catch up to technological change; (3) Companies that become Monopolies rarely stay that way, especially companies that have extensive regulatory oversight like the U.S. Exchanges do; (4) customers and competitors of the Exchanges have begun a unified and significant public campaign challenging this fact and (5) the SEC, one of the primary regulators of the Exchanges, has recently named regulatory reform of the Exchanges as one of its most significant priorities.

In short, as displayed in more detail below, when the financial data, public statements by the CEOs of the Exchanges, sell-side analysts and SEC Commissioners all publicly state that the exchanges have become a monopolies, it's unlikely they stay monopolies and investors should be wary that the earnings power they've ascribed to the companies will continue unabated, especially when this is a rare instance where there has been bi-partisan support across both political parties to begin extracting change.

The likely result of these actions, or even the market's realization that the threat that these actions may occur, will be to lower the earnings power of the Exchanges, downgrade their multiple, and inject significant uncertainty into the stocks driving their valuations down considerably. Or to think about it inversely, it's highly unlikely that the combination of these effects can increase earnings power or the multiple of the Exchanges, making the upside reward of owning these stocks significantly disproportionate to the downside risk.

We have seen this story play out multiple times: we saw this occur in the pharmaceutical industry just three years ago when it was uncovered that they had been raising drug prices for years on unsuspecting patients. The pharmaceutical index returned 188% from 2011 to 2015 as investors watched rising drug prices drive significant earnings power, only to have unified customer outcry significantly effect the pharmaceutical companies' earnings power and valuation multiple; as a result this same index has decreased 25%, underperforming the S&P by over 85%. We saw it again this year, most recently and prominently with Facebook, where not even has a single regulation been enacted; there has not even been a specific regulatory proposal made. But the mere threat of regulatory reform inevitably causes significant uncertainty, invariably driving down multiples and stock valuations. Even with absolutely no regulatory reform enacted, Facebook's stock has decreased 25% since July. In situations of uncertain regulatory reform and public outcry, we've learned companies always follow the same playbook: They self-regulate to appease the regulators and influence public opinion; this self-regulation often takes the form of permanent changes to the company's business model to structurally lower profits proactively addressing concerns. For example, following the threat of negative press and regulatory scrutiny, the largest pharmaceutical companies in the world all responded the same way: Novartis, Pfizer, Merck, Sanofi and Roche all independently announced they would no longer raise drug prices (thereby lowering their earnings and multiples) and as described, valuations dropped significantly causing the companies to significantly underperform their prior performance. Similarly, in order to be proactive and get ahead of regulatory scrutiny, we saw Facebook follow suit: announcing voluntarily actions to increase disclosure ("we took major steps to make advertising...more transparent."), make significant downward revisions to their revenue guide causing revenue to decline year over year for the first time in their history, partly to give increased data privacy choices to customers, and perhaps most importantly, guiding to a major decrease in their long-term profitability lowering consensus margins nearly 10 points. Facebook's stock declined 19% the next day and has continued to fall, dropping another 8% over the past two months. What should be obvious is the Exchanges will follow the same self-regulating playbook – in the face of what we believe is significant pressure, which we go into detail below, we expect the Monopoly model that has driven a 20% EPS CAGR to cease to exist as they take steps to address the biggest concerns - for example, we would expect their next earnings guide to remove the 30% price increase planned on customers, in addition to other reforms that have been aiding their inflated guidance in order to get in front of increased scrutiny. Investors can play with various scenarios, but to help frame risk, even a 10% decrease in earnings guidance and just a one or two multiple correction can easily result in the 20%-25% price declines that we saw occur in similar situations.

### Details

For purposes of this writeup, we'll use ICE's earnings to illustrate our points, though as stated above, the crux of the short thesis applies generally to the US exchanges as a whole.

ICE's revenue is approximately broken down as 51% Data & Listings and 49% Trading/Clearing. The exchanges are trading at or slightly above their historical trading range with consensus estimates implying the Exchanges outsized monopoly profits continue in perpetuity. Below are the forward p/e ratios over the past 5 years of the exchanges:

Today, those multiples remain roughly intact, i.e. ICE continues to trade 20x their forward P/E while recently making their high stock price of the year.

The US exchanges should not be trading in-line with their historical value, given significantly increased regulatory risk that has transparently been signaled to be a near-term event. Historically, exchanges were not-for-profit organizations formed to oversee trading of the public markets. In the last decade, this changed where they have become publicly traded corporations whose primary function has changed from equitable oversight of the public markets to one designed to maximize profits. Further, recent consolidation among these companies mean that while we have 13 public stock exchanges, 12 of them are now owned by just three companies. This is a large reason why the exchanges have begun to have the ability exert unusual and undue pricing power over their customers and develop monopolistic tendencies that have become increasingly questioned by a broad swath of market participants across the political spectrum, buy and sell side firms, brokerage firms, journalists, customers and even federal courts themselves – a universal consensus from market participants with varying financial and political incentives not usually seen. The SEC has come under increasing and universal criticism as failing to adequately oversee Exchanges and ensuring regulations keep pace with technological change in the markets. This criticism has understandably drawn the attention, and embarrassment, of the SEC, and as we demonstrate definitively below, we believe they have taken a renewed, intense and immediate focus on prioritizing regulatory reform of these exchanges that will have serious ramifications to their business model.

We'll begin with a brief overview of the company's financials and commentary indicating their increasing pricing/monopoly power from the US Exchanges over market participants and the indications from their largest clients and the SEC that they blatantly recognize this issue and have displayed a clear intention to begin regulatory reform. We'll also address why this has become a pressing issue now and therefore makes the trade timely today versus sometime in the past or future, thereby indicating near-term action is likely, as well as a number of leading signals that further verify the likelihood of action. Finally, we'll address the counter narrative the companies have begun to push to sell side analysts and investors to protect their stock prices and attempt to minimize concerns, and why we believe these narratives are naïve or lazy as to their practical effects on the business and stock price, and how we think about potential regulatory reform on the business.

### **Monopoly Power**

Monopolies are defined as an advantage obtained by an entity or entities over the commercial market in a specific area. Common elements that signal monopoly power exists is pricing power over customers (the ability to raise prices on customers and their inability to resist these increases or go to a cheaper competitor), especially when there are limited suppliers of certain products or data or an ability to exclude competitors.

ICE has grown its Data business from 12% of revenues in 2012 to over half the business today. In their 2017 analyst day presentation the previous 10 year period showed Data revenues grew from \$35mm to \$2b, an incredible 50% CAGR. Going forward, they continue to guide 6%-7% growth in this business. Of this growth, they explicitly guide that 30% of this growth will come from increasing prices on their customers. There are many factors we think of when we think of pricing power; but a large number of ICE's customers are some of the biggest companies in the world: for example, JPMorgan has a market cap of nearly \$400b; Citi's market cap is nearly \$200b, Goldman Sachs, Morgan Stanley and Blackrock have market caps between \$75b-\$100b. These are not ordinarily the type of companies that we think of as being unable to negotiate with their suppliers or are forced to eat price increases, yet not only have they been doing so all decade, they have been guided that they will continue to do so. In fact, it appears the pricing power is strong as ever. Just a few months ago on their Q1'18 earnings call, ICE's CEO stated, "We've been very open...with our customers that we think we can grow this data market at high to mid-single digits. We think it's going to grow like that for quite a while...and when we sit across the table and negotiate we basically say to people this is the kind of return that we're looking for if you want to do business with us... it's really been successful. We've really done a very good job, getting 11 out of the 13 top managers onto this platform."

This is fairly incredible – it's not often a company sits across the table from the 13 largest managers in the world, tells the those managers that if they want ICE's business, they'll take the price dictated to them, and have that strategy result in (some may say monopolistically) getting 85% of the largest customers to not only sign and become customers, but to take the price dictated to them. And then it is really a special kind of pricing power that allows the CEO to follow that up with publicly going on the next earnings call to tell the world that's how it works "if [they] want to do business with us".

There are few reasons these monopoly conditions exist; a combination of consolidation, lax SEC regulation, historical rules and significantly, an interesting interplay between firms needing to follow SEC regulations to be SEC compliant and the exchanges being the only companies that can provide them with the systems and data needed to do so. In a research note titled "Not all Market Data...are created Equal", UBS describes these exact monopolistic conditions for the exchanges, stating, "It also helps to have a must-have product or set of data, such as a virtual monopoly on proprietary market data [such as ICE or NDAQ]...We believe ICE and NDAQ's unique access to data generated by operating exchanges puts them in an enviable position to provide must-have data [to their customers]".

The CEO of ICE himself has alluded to the same fact pattern: in November 2015, SEC/FINRA released a Regulatory Compliance Release updating its Best Execution rules where they instruct firms they need to continue to have the most updated technology to best comply with trading rules (then using regulatory releases like these, the exchanges sell the "new and best" data and "proprietary technology" to their customers). Until this release, much of the SEC's regulatory focus was on equity markets. This particular release, beginning on page 7 titled "Additional Considerations for Best Execution for Fixed Income Securities", the SEC acknowledged in the first sentence that the fixed income market "has evolved significantly in recent

years” and goes on to say that a firm must take into account market and technology changes that might alter its best execution analysis. The SEC recognized that fixed income securities do not have as transparent pricing as equities, and as a result made a number of recommendations firms should follow to be compliant with their Best Execution rules in fixed income markets. Coincidentally (or, not), just one month after this SEC release describing new best execution rules for the fixed income market, ICE purchased IDC in December 2015 for over \$5b, a company that specializes in “pricing and valuation on fixed income securities”. Of course, combining IDC’s services with the SEC’s November updated notice makes the sales pitch to customers quite easy, and of course they will have to pay almost any price you charge them in order to remain SEC compliant. In fact, the CEO went ahead and said so himself. On the very first earnings call after purchasing IDC (4Q’16, Feb 2016), the CEO begins by validating that the value of the business comes from mandatory regulatory reform, and follows that up by stating that promptly upon purchasing IDC, they immediately raised prices on their customers. From the aforementioned earnings transcript:

Page 8: CEO of ICE: “In my mind, a lot of the opportunities set that we’re seeing really comes from the fact that the buy side was able to rely on certain services that were relatively informal. But today, due to regulatory reform and internal audit practices, need to be more rigorous. [and] the regulators and the auditors want an independent validation, increasingly, as people are taking the obligation to get best execution seriously as the SEC is paying more attention to best desk requirements in the fixed-income space, asset managers and others are [forced to] make investments to make sure that they are compliant.”

Regarding their pricing power, the CEO followed that up with: “What we’ve always seen in data is that...people are happy to pay more for it. And so, in that regard, we don’t get a lot of pushback on our [increased] pricing. And yet, we’ve been able to really raise the profitability of these businesses. The AllianceBernstein product that I mentioned in the prepared remarks is [another] deployment that results in financial gain for us.”

Further, the economic evidence also supports this thesis: in an article by Forbes last week describing a speech SEC Commissioner Jackson’s speech gave, they say, “Jackson complained the exchanges have morphed into monopolies—with monopoly profits to match. Of the 13 stock exchanges, 12 are owned by three corporations: the New York Stock Exchange, Nasdaq and Cboe. Since it went public in 2006, the NYSE’s parent, the Intercontinental Exchange (ICE) has had total annual returns of 24.1%. He continued on to say, “Its 54.3% net profit margin last year was the fourth highest of any company in the S&P 500.”

(<https://www.forbes.com/sites/tedknutson/2018/09/19/stock-exchange-reforms-needed-to-save-investors-money-urges-sec-commissioner-jackson/#18807a947d12> )

Additionally, we’ve mentioned that we have 13 exchanges with 12 of them owned by three companies. One might ask why do we have so many exchanges, if they are owned by only three companies? We understand why companies acquire competitors with the same business model. It becomes harder to understand what would drive a company to acquire, and then to continue to operate, virtually similar businesses unchanged. The answer, of course, is that they charge investors to connect to each exchange individually, incentivizing them to keep them all. You may also recall the SEC’s embarrassment at allowing the Exchanges to become monopolies, which in turn is partly why they are urgently motivated to reform them. It turns out that regulations state the exchanges have to ask the SEC when they want to increase the fees they charge to their customers for connecting to the exchanges. Note that the cost to the exchanges for connecting to electronic connections drop every year. Yet despite these declining costs, we have counted an astounding 95 times in the past three years the exchanges have asked the SEC to increase the price they charge to their customers. The SEC granted the price increase all 95 times. I should note that just last week, they asked the SEC a 96<sup>th</sup> time to increase prices: and for the first time this SEC said no, a leading indicator for what’s to follow.

We’ve shown that the financial data, CEOs of the Exchanges, the regulators and sell side analysts all believe these companies are monopolies. Below we will display further examples from Congress and other market participants. But in a remarkable indication of overwhelming consensus, even the White House has reached exactly the same conclusion.

Treasury Secretary Mnuchin released a comprehensive report on his assessment of capital markets last year titled “A Financial System...”. In the section of the report aptly named “Market Data”, he specifically says “Many broker-dealers report that they feel compelled to purchase these enhanced data feeds from the trading venues both to provide competitive...services...and to meet their best execution obligations. Broker-dealers interpret their best execution obligations as requiring them to use the best available data...[as a result the] firms [are forced to] employ proprietary data feeds to identify the best prices...In addition, the market for proprietary data feeds is not fully competitive...[therefore] limited constraints on exchange pricing power has allowed exchanges to regularly raise prices. Consequently, exchange data fees made up nearly a third of exchanges’ \$28.3 billion in 2016.” In this same report, the Treasury department explicitly recommended that the SEC issue guidance that broker-dealers do not necessarily have to subscribe to proprietary data feeds. He goes on to state “This should help to eliminate the need for broker-dealers to defensively subscribe to these costly data feeds to ensure that they meet...their best execution obligations...Treasury [also] recommends that’s the SEC also recognize that the markets for...data feeds are not fully competitive. Perhaps most importantly, in direct contradiction to the Exchanges attempt to downplay the SEC’s authority to effect pricing, Treasury Secretary Mnuchin goes on to explicitly state that the SEC has the regulatory authority to address these competitive concerns saying: “The SEC has the authority under the Exchange Act to determine whether fees charged by an exclusive processor for market information are “fair and reasonable” [and] “not unreasonably discriminatory” when determining whether to approve...data fees”.

### **Customer Reaction**

Given the Exchanges monopoly power, one wouldn’t be surprised that their customers are unhappy. One has to look no further than Customers’

recent reaction to the Exchange's price increases. As a further indication that public pressure will increase on the Exchanges, the customers are no longer remaining silent, publicly declaring their displeasure and demanding reform as we display below.

Instead of JPMorgan, Goldman Sachs, etc using their market power to lower prices as they do other suppliers, as we detail below, they instead delegated their top executives to take significant multi-million dollar cuts from their salaries at these banks to temporarily become top executives at the SEC to exert regulatory influence in order to have reforms enacted to remove this monopoly pricing power from the Exchanges (and therefore lower Exchanges earnings power). So these customers didn't threaten to go elsewhere; they didn't threaten to use a different part of their business; they didn't create the data themselves; they didn't even get together with the other banks like they did when they created Zelle to compete against Venmo – they have been eating price increases from this magical pricing power the exchanges have, and unable to exert pricing power back, they have resorted to having their top executives move to the SEC for one goal only: to reform the Exchanges. This, along with the majority of the SEC's backing and significant bi-partisan support, is why reform, and the noise around regulatory reform, has begun to amplify.

There are a number of examples of the above, but here are just two:

Brett Redfearn, who was the head of all Market Structure at JPMorgan, recently left JPMorgan to become the Head of the SEC's trading and market division. (<https://www.bloomberg.com/news/articles/2017-10-18/a-jpmorgan-executive-is-said-poised-to-take-key-role-at-sec>). As the article states, Redfearn's main goal as head of the SEC's trading & market division is to reform the exchanges and lower prices. Quoting from the above article:

**“Redfearn will likely have significant sway at the agency because of his expertise.** Redfearn, a JPMorgan employee for more than nine years, expanded his role at the bank earlier this year. He went from just overseeing equity market structure strategy to running market structure for all asset classes. He's advocated for a regulatory overhaul. In April, he said Regulation NMS -- a landmark SEC rule approved in 2005 that accelerated a shift to electronic trading in the U.S. stock market -- is **“overdue for reform.”** **The pick may set up a clash with stock exchanges,** who are among the most influential voices in Washington around financial regulation. **Redfearn has emerged as a critic of the increasingly costly fees that U.S. stock exchanges charge** traders who want access to vital data on prices.”

Here's another example from Goldman Sachs. Note first that the Chairman of the SEC has had an extensive relationship with Goldman Sachs as his client prior to joining the SEC. He has just recently hired a former Goldman Sachs executive as his Senior Policy Advisor to address these same issues. <http://www.nydailynews.com/news/politics/sec-chairman-hires-goldman-sachs-exec-senior-policy-advisor-article-1.3461141>

Note that this same article also discusses the hiring of the SEC's newest commissioner, who has been leading the charge in taking the Exchanges head on.

SEC Commissioner Jackson last week gave a significant policy address, titled, “Rethinking Exchange Regulation” where he specifically addresses all the points made above. The reaction to this speech by customers was applauded, rapid and swift:

For example, JPMorgan wrote a letter to the SEC supporting reform. Business Insider wrote about their letter, stating, “For example, an especially important aspect of our proposal was to include a “no-rebate” bucket that will allow us —and, more importantly, investors — to observe how markets respond to the absence of rebates,” Jackson said in a speech at George Mason University in Virginia.

One market observer said JPMorgan's submission of a letter “is like an 800 pound gorilla wading in.” “Plus they're a NYSE issuer breaking ranks,” the person added”

Similarly, within hours of SEC Commissioner Jackson's speech, SIFMA, the Securities Industry's largest trade group representing 75% of all broker dealers, released their own statement explicitly commending Commissioner Jackson's recommendations on these issues: (<http://www.mondovisione.com/media-and-resources/news/sifma-statement-on-sec-commissioner-robert-jacksons-call-for-exchange-reform/>)

### **Congressional Reform and Bi-Partisan Support**

In SEC Commissioner Jackson's speech, he also specifically addressed bi-partisan support for reform of the Exchanges, stating, ““This is not a partisan issue: Commissioners of all stripes, Members of Congress and our current Chairman have all said it is time for us to take a hard look at the structure of our stock markets. We have a rare bipartisan opportunity to move this forward—and, for reasons I'll explain, we owe it to American investors to take it.”

The Chairman of the SEC, Jay Clayton, in a speech laying out the SEC's priorities to the Economic Club of New York stated, “I am also pleased to note that this week, Chairman Hensarling and Chairman Huizenga of the House Financial Services Committee and its subcommittee on Capital Markets, Securities, and Investment...I look forward to working with Congress on these issues.”

<https://www.sec.gov/news/speech/remarks-economic-club-new-york>

Of course, as laid out above, Treasury Secretary Mnuchin also backs reform. Additionally, Congress has begun to actively focus on this issue as they are starting to take a progressively more aggressive role. Both Democratic and Republican Senators have begun to write letters to the SEC highlighting the need to reform the Exchanges. To cite just a few examples:

Senator Charles Schumer addressed a letter to the SEC Chairman arguing that exchange rebate structures “create a conflict of interest, as brokers may be incentivized to execute trades on a particular venue even if that venue is offering the best price”); *see also*

Republican Senator Michael Crapo and Democratic Senator Mark Warner also wrote a letter to the Chairman of the SEC, addressing serious concern about the SEC's "pace of reform efforts" in addressing market structure

Senator Carl Levin wrote a letter stating that "Conflicts of interest erode public confidence in the markets and have the potential to harm investors and I believe the SEC should take prompt action to eliminate these conflicts of interest."

Senator Edward Kaufman addressed the issue as well, writing that "[S]everal areas of current market structure lead me to be concerned about the performance of the markets for investors and companies seeking to raise capital."

### **Recent Actions Signaling Prompt Regulatory Action**

Wasting no time, and demonstrating the intense and near-term focus on this issue, just last week, the SEC announced an all-encompassing round-table specifically addressing Exchange Reform titled, **"SEC Staff to Host Roundtable on Market Data and Market Access"** designed specifically around reforming the Exchanges:

<https://www.sec.gov/news/press-release/2018-210>

### **News Coverage around Regulatory Reform have begun to increase, reinforcing incentives for Regulator to Act**

Just in the past week, both the Financial Times and the Wall Street Journal have written numerous articles describing both the need to reform the Exchanges business model and the SEC's desire for major reform and their urgency on the issue.

The WSJ, in an article titled, **"Brokers Notch Wins in Fight Over NYSE, Nasdaq Data Practices. Regulators step up scrutiny of stock-price feeds sold by exchanges"** describe that:

"Banks and brokerages are starting to win more battles with big stock-market operators over the cost of data, threatening a lucrative revenue stream for the New York Stock Exchange and Nasdaq Inc. The Securities and Exchange Commission has recently ratcheted up scrutiny of the stock-price feeds sold by exchanges..."

The SEC also last week suspended price increases—which would have become effective without the agency's intervention—proposed by three options exchanges...

Industry executives credit SEC Chairman Jay Clayton and a new director of trading and markets, Brett Redfearn, for the [stepped-up scrutiny](#) of exchange fees. SEC Commissioner Robert Jackson Jr. said last week it was "time for the SEC to get serious about that obligation."

<https://www.wsj.com/articles/brokers-notch-wins-in-fight-over-exchanges-data-practices-1538049600>

The FT wrote an article this week titled, **"Battle intensifies over the costs of using US Market Data"**.

They state, "At issue is whether the prices are fair and reasonable, which securities regulations dictate they must be. In the middle is the Securities and Exchange Commission, which has the power to suspend fees. Last month, Robert Jackson, an SEC commissioner, [launched](#) a broadside against the for-profit exchanges that dominate US markets."

The article also says that the "The SEC has received [petitions](#) calling on the regulator to examine the status quo for disseminating market data. One [was signed](#) by some 24 companies spanning the equity industry including Citadel Securities, a market maker, as well as Fidelity and Vanguard, which are both asset managers."

<https://amp.ft.com/content/66ad0f84-9970-11e8-88de-49c908b1f264#>

Further, influential trade magazine Pensions & Investments published the following piece:

<http://www.pionline.com/article/20181001/PRINT/181009977/regulators-to-take-close-look-at-power-of-exchanges>

### **Exchanges Response**

The U.S. Exchanges of course cannot debate the intense and picking up regulatory scrutiny of their practices, and have attempted a counter-narrative stating (1) that the regulation only effects equity markets which they attempt to define as effecting only 9%-10% of their revenue; (2) they state that regulatory reform may take some time to complete, so don't worry about it now; and (3) denying there is enough support from regulators to extract regulatory reform and citing that beyond the SEC other regulators (such as the CTFC) also regulate the exchanges.

We believe we've addressed most of these points as incorrect, but will briefly address them again here. First, potential regulation is clearly not restricted to equity markets. Commissioner Jackson in a separate speech in July specifically focused on the importance of fixed income market reform as part of overall regulatory reform, which encompasses a far greater portion of revenues than the Exchanges would have you believe. For example, he states, "Despite the progress in our stock markets, fixed-income markets—where millions of Americans are now shifting their investments as they prepare for retirement—remain rife with conflicts of interests and hidden costs. For years, Members of the Commission and finance scholars alike have worried that fixed-income markets do not provide the level playing field that American investors deserve, but we have not yet moved forward with reform in that area...Because we've invested so much time and Commission energy in our stock markets over the past decade, the investor benefits of turning our attention to fixed-income markets would be significant. That's why it's so important that today's rule includes a clear commitment to review our approach to fixed-income electronic trading platforms."

Secondly, "regulation takes a long time" should be a worrisome answer. It implicitly admits there's an issue; that the investor should perhaps only own the stock until closer to regulation; it ignores all empirical evidence above around the urgency of the issue, and most importantly, as discussed above public scrutiny and potential regulatory reform generally not only cause structural business changes effecting earnings power and valuation, but there are numerous examples of the threat of regulatory reform effecting the stock price. We find this answer inadequate.

Finally, an argument that investors shouldn't only worry about the SEC because there are other regulators we believe not only understates the SEC's influence, but demonstrates a misunderstanding of regulatory reform. We have already discussed that both chambers of Congress have written letters addressing this issue, significantly broadening the potential breadth and outcomes of regulation. Further, despite the Exchange's narrative of the limited scope of the SEC's regulation, Treasury Mnuchin himself stated "The SEC has the authority under the Exchange Act to determine whether the fees charged by an exclusive processor for market information are 'fair and reasonable'" which would significantly broaden this bucket. We also believe this demonstrates a misunderstanding of how regulation works. Regulators are not boxed off but constantly talking to customers and assessing data; and those market customers are unhappy and the data is unfavorable to the exchanges. But despite all of this, even the White House has asked other regulators to coordinate in reform. In the report cited above, Treasury Secretary Mnuchin explicitly calls for the CTFC to coordinate with the SEC on regulatory reform, further negating the Exchanges argument around a lack of coordination among regulatory agencies.

As Reuters describes the report, "Christopher Giancarlo, chairman of the Commodity Futures Trading Commission (CFTC) and Jay Clayton, chairman of the Securities and Exchange Commission (SEC), said in statements on Friday they had provided extensive input to the "thoughtful" report and supported its recommendations...The Treasury also waded into the long-running debate over equity market structure, proposing the SEC review share-tick sizes, order types, exchange fee models, and how exchanges themselves operate and are governed.

To reduce regulatory duplication and bring the United States more in line with other markets, the report calls for the SEC and CFTC to work more closely and harmonize their rules, but stopped short of recommending a merger of the two - something policymakers have called for in the past.

...Friday's report, in contrast, outlines a broad range of 91 technical fixes aimed at boosting stock, bond and derivatives markets. All but nine can be put into effect by the country's federal regulatory agencies, who were consulted on the report.

The recommendations were met with quick praise from financial industry groups, who said capital markets regulations were also long overdue for a review."

<https://www.reuters.com/article/us-usa-treasury-regulations/u-s-treasury-outlines-sweeping-reform-of-capital-markets-idUSKBN1CB1RD>

## Other Risks

While we have addressed some of the most pressing risks facing the Exchanges, we'd be remiss to not at least mention that there are a number of other areas where investors are underappreciating potential risks or changes to the business model. Our main point of this section is less about likelihood of change or materiality to the business, and only that the business model is one that is under scrutiny in many different areas, making it much more unpredictable than investors believe, with unknown effects to how it could affect their earning power.

Just to cite a few examples:

One example is that the maker-taker rebate model used by exchanges is under much scrutiny. We won't get into the details here, but suffice to say there is significant concern that rebates are being used as way to route orders to certain exchanges, enhancing their market power. The CIO of Yale's endowment described it this way, <https://www.nytimes.com/2017/07/18/opinion/wall-street-brokers-rebates-kickbacks.html>

Second, in addition to being regulated by a number of government agencies such as the CTFC and SEC, they are also designated a SRO (self-regulatory organization) which provides them certain benefits. For example, they are able to play role in helping to design the rules they are regulated

by. Further, their 10-K discusses certain possible immunity benefits stating “[...] maintain[s] our exchanges’ abilities to operate as SROs with related immunity for the discharge of their regulatory functions.” As previously described, ICE’s business has transformed over the past decade, such that less than half their revenue is transactional, or directly related to trading vs. the Data & Listing business. They also announced the acquisition of loan registry company MERS as they enter the mortgage market. While ICE began as a non-for-profit marketplace exchange, they are now a vastly different company operating in the mortgage market and generating the majority of revenue via non-transactional means, such as data. While that data is certainly market related; they are arguably less of the traditional exchange and more of Data company, blurring the lines between recurring software companies, data feed companies and mortgage markets, with less than half their revenues being transaction based. To a degree this is occurring across the industry, though the rate of change, types of industries and monopoly power opens up questions about how the companies should be expanding and if there is risk being added to the system. It’s certainly far from clear the SEC continues to grant them SRO status at renewal.

Third, currently the exchanges operate somewhat immune of liability. Under a legal loophole, they assert they are immune from liability on the theory they are acting as regulators rather than profit-makers. They also protect themselves by requiring all traders to agree to liability limits, generally a maximum of \$6 million. For example, as investors may recall, Nasdaq was negligent during the Facebook IPO, having technology glitches that delayed the opening and left thousands of orders stuck in the system for hours. Estimates put losses from market-making firms at \$500mm due to Nasdaq’s errors. The SEC has proposed to eliminate this rule. While such glitches are rare, it exposes an additional tail risk to the business model.

### **Trading and Risk/Reward**

Even if one continued to remain unconvinced of the above, a large percentage of investors hedge their portfolio with the S&P index against their individual long stocks. The correlation between the exchanges and the S&P is 0.94. With no near-term upside catalysts (i.e. earnings), the exchanges could be viewed as a portfolio hedge, where investors replace portions of their index hedge, retaining optionality on increased newsflow and regulatory reform to retain the ability to earn outsized returns on a short position, with limited upside risk. We believe this allows an investor to size the shorts at a bigger size as they act as a portfolio hedge while the thesis does or doesn’t play itself out over the coming weeks.

Liquidity: All the exchanges trade with plenty of liquidity and extremely low current short interests, ensuring trading technicals also rest in the short investors favor.

I do not hold a position with the issuer such as employment, directorship, or consultancy.

I and/or others I advise hold a material investment in the issuer's securities.

## **Catalyst**

Congressional Reaction, White House Reaction, Regulatory Reaction, SEC, CTFC, Reduced Guidance



## Messages

|                |                     |
|----------------|---------------------|
| <b>Subject</b> | Another Short Risk  |
| <b>Entry</b>   | 10/10/2018 07:39 AM |
| <b>Member</b>  | VIC_Member2015      |

Another major risk we failed to mention that we believe investors' do not understand but is a substantial liability is the upcoming ruling in the SIFMA case that could cause earnings to be cut substantially. As described in last week's WSJ article, the SEC Commissioners are expected to rule this month, and the stakes are high: <https://www.wsj.com/articles/brokers-notch-wins-in-fight-over-exchanges-data-practices-1538049600>

As described in our writeup, one reason this has become a major issue for the SEC is that they been criticized for allowing the Exchanges to become a monopoly, which has caused all five Commissioners to make this a top priority. For example, recently a federal appellate court said the following:

"A federal appeals court on Tuesday criticized the Securities and Exchange Commission for approving a fee plan by the options industry's major clearinghouse and told the regulator to redo its work. A three-judge panel of the U.S. Court of Appeals for the D.C. Circuit wrote that the SEC "effectively abdicated" its responsibility to review Options Clearing Corp.'s plan..."

[https://www.wsj.com/articles/appeals-court-orders-sec-to-reconsider-options-industry-fee-plan-1502213910?mod=article\\_inline](https://www.wsj.com/articles/appeals-court-orders-sec-to-reconsider-options-industry-fee-plan-1502213910?mod=article_inline)

More importantly, we also described that SIFMA, a lobbying group representing 75% of all broker-dealers, strongly commended Commissioner Jackson's recent speech where he called for major regulation of the Exchanges and SIFMA recognized early on that the Exchanges had become monopolies, agreeing with SEC Commissioner Jackson.

In fact, they filed a lawsuit against the Exchanges for using their monopolistic pricing power to take exorbitant price increases; a decision that has now come to a head as the SEC will rule in the coming weeks as to whether to rule that the Exchanges price increases were unlawful. SEC Commissioner Jackson recently described the Exchanges monopolistic pricing power in his September 19<sup>th</sup> speech and is scheduled to address it again tomorrow October 11<sup>th</sup> at a conference specifically addressing Monopolies. This case has extraordinary ramifications for the industry as it has enjoyed lax unchecked regulation for years. We think there is significant analyst and investor complacency given the decision's potential effects.

We have described in detail the excessive price increases the Exchanges have taken on the "data" portion of their business and their unabashed guidance to continue to such behavior. For example, R2G, a data-based services firm, estimates that the three largest exchanges (ICE, NDAQ and CBOE) grew revenues for their "premium" data by 253% from 2012 to 2017. For example, in 2012, the monthly cost for such data was ~\$72k. In 2017, that same monthly cost was ~\$183k.

As we have described, this is only a small, but representative example of the Exchanges pricing power and extravagant pricing behavior. We believe part of the analyst and investor complacency comes from running rough calculations on what percent of revenue an item like this represents for Exchanges, classifying as "at risk" and becoming falsely comfortable with what they believe downside might be. But as we note, this is only a tiny piece of what is classified as "Data" revenue, which as a portion of overall revenue represents over half of their actual revenue.

As discussed above, the Exchanges consistently push a robotic narrative to analyst and investors alike that the SEC only has the ability to regulate a very small portion of their fees - for example, ICE frequently cites that the SEC only has the ability to regulate approximately 5% of their revenue so investors need not think about it. In actuality, SEC regulations cite that fees must be "fair and reasonable", among reviewing a number of conditions, including how competitive the market is, which in fact gives the SEC precisely the jurisdiction needed to review these monopolistic price increases.

[<https://www.sec.gov/alj/aljdec/2016/id1015bpm.pdf>]

There are a remarkable number of egregious violations: For example, prior to 2017, "Network A" data cost \$270 for use on ten devices. In 2017, the NYSE increased the fee for the same data to \$6,000. See: <https://www.sec.gov/comments/sr-ctacq-2017-02/ctacq201702-1710463-150167.pdf>. On March 31, 2013, NYSE "Openbook" cost \$14,000. The following day, that same product cost \$21,000. <https://www.reuters.com/article/sec-exchanges-netcoalition-idUSL2N0DH1TF20130430>

There are numerous examples of increases like this, which helps explain the incredible growth and monopoly pricing power the Exchanges exhibit. The point here though is not to demonstrate price increases or monopolistic pricing power, which we've already established above. The point is to signal the importance of this specific case that the SEC will rule on in the coming weeks. If the SEC rules that the Exchanges have not been taking "fair and reasonable" price increases as securities regulations clearly dictate (contrary to what the Exchanges would have you believe), the result would be to roll back many of these price increases, many going back as far back as 8 years, as well as severely prohibit such pricing attempts in the future, resulting not only in a slowing of the Exchanges growth rate, but would substantially cut existing revenue from previous unreasonable price increases they have already illegally been taken (and margin, and profits).

Further, a secondary clear read-through is that a vote by the Commissioners in SIFMA's favor would plainly indicate where they stand on the Exchanges becoming monopolies, opening up increased regulation and sending a signal to other potential Monopolies. Note that less than two decades ago these



companies were not-for-profit exchanges designed to operate in investors' best interest versus for-profit entities designed to maximize shareholder returns. We believe investors not only severely underestimate the importance of this case, many are unaware the case exists. In our conversations with analysts and investors, regulatory concerns are consistently naively dismissed by repeating "revenue at risk" percentages, conveniently dictated by the Exchanges themselves, but without citations to the underlying SEC regulations [<https://www.sec.gov/alj/aljdec/2016/id1015bpm.pdf>].

While the Exchanges are not Valeant, the analysis pattern is remarkably similar. When Philidor was first discovered, investors attempted to quantify downside to stay comfortable with their investment, believing only X% of revenues were at risk without further analysis. We think this upcoming case is possibly a watershed moment for the Exchanges, or atleast has the potential to be and certainly the way the SEC Commissioners vote should tell investors the likelihood of regulation. We consistently find it surprising that investors believe that our U.S. public stock and fixed income markets, consistently cited as the most liquid, reliable and important such exchanges in the world, believe a narrative that they are unregulated or that "only 5% of revenues are regulated".

A vote by the Commissioners in SIFMA's favor should also have a significant secondary effect as it indicates to investors their stance on the years of under-regulation the Exchanges have enjoyed. Pricing examples like the above likely also open up the Exchanges to criticism by other regulatory bodies as they begin to pull on the thread to begin to examine other areas of their business that have come under criticism, such as the use of rebates to drive business toward existing exchanges, arguably contributing to these monopolies' existence in the first place. (See, e.g. "[Rebate] payment to brokers when not shared with the broker's client is equivalent to a kickback. Public data shows that exchanges who pay this rebate garner a greater percentage of order flow despite providing worse execution quality." Brad Katsuyama, CEO of IEX, House of Representatives Testimony)

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|----------------|------------------------|
| <b>Subject</b> | Re: Another Short Risk |
| <b>Entry</b>   | 10/25/2018 04:28 PM    |
| <b>Member</b>  | SpocksBrainX           |

again, thanks for your fine work

I read the NDAQ call - the CEO was livid, and made some good points (namely, that the 400 filings sent back was a dump considering some of those involved prices decreases and no increases, so there was some outright piling on to make a point, rather than make sense - this is my take). And it seemed obvious that they aren't going to placidly accept things, with legal actions coming and endless litigation in the hopper perhaps (and any time limit to this stuff is an absolute joke). Course, you made a very going point here - at the least, this is going to enrich a lot of lawyers and maybe cause absolutely endless litigation, and at the worst this crud will cast a pall over these stocks for the next who knows how many years. I also thought it was weird that maybe this is coming down party lines (if I read this right) - democrats in favor of socialism and big banks (odd combo), republicans in favor of capitalism, but maybe that's outright cynicism on my part (though I mean it literally - if you don't believe me, read the call). All that said, unless there are other motives here - a Morn analysts suggested this was a gambit to cause big banks and the exchanges to hash out some deal which will no doubt leave the little guy completely out of it - it sure doesn't read like good news, though for the life of me I can't - and nobody in that call could - actually quantify any of the risks, including the companies themselves. My great non-informed opinion is that the risk is nowhere near as bad as it can sound, and that cooler heads will eventually prevail, but there is no way to know. Regardless, the exchanges aren't particularly cheap here and again I want to thank you for bringing this up. I remain unsophisticated, but the one thing I hate is utter uncertainty, and having a regulatory body determined to scr\*w with you doesn't seem like a good thing period (and messing with somebody means that you agree to something and then change your mind several years later with a regime change - what a true cluster and pretty much what you expect whenever government is involved in darn near anything).

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*So I think it's really important to read the statements from the Chairman and from the Republican Commissioners. I think the first thing to note is that they're not saying that, necessarily, that our fees are not reasonable. I mean, it's kind of a double negative. But they are saying that they would like more evidence to support the fees that we are -- have delivered. What's frustrating to us is that every single one of those fees is approved by the SEC, and so they've all been reviewed. At no point did they ask for further information before granting us the approval. So now they're coming back and saying, "We want more information." And we have to understand that in terms of what they want. It's very -- also very clear that the Republican commissioners have said they're not looking to rate-make, they're not looking for a cost-plus model, they're just looking for further justification in the context of it being at a competitive market, having competition around the Market Data.*

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and watch out everybody else - they are coming for you next

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|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Subject</b> | <a href="https://www.wsj.com/articles/wall-street-firms-plan-new-exchange-to-challenge-nyse-nasdaq-1154686612">https://www.wsj.com/articles/wall-street-firms-plan-new-exchange-to-challenge-nyse-nasdaq-1154686612</a> |
| <b>Entry</b>   | 01/07/2019 04:03 PM                                                                                                                                                                                                     |
| <b>Member</b>  | pcm983                                                                                                                                                                                                                  |

[https://www.wsj.com/articles/wall-street-firms-plan-new-exchange-to-challenge-nyse-nasdaq-1154686612?mod=hp\\_lista\\_pos2](https://www.wsj.com/articles/wall-street-firms-plan-new-exchange-to-challenge-nyse-nasdaq-1154686612?mod=hp_lista_pos2)